

MATH 3551 A — Abstract Algebra I

Additional Practice Midterm 1 — Version C

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For practice — Midterm 1 is Monday, October 12, 2026 • Time limit: 50 minutes • Total: 100 points

NAME (please print): _____

Directions. This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem, $|H|$ divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

About Version C. This additional practice exam was generated with GPT-5.6 Sol in Codex. It uses the current Midterm 1 coverage and targets about thirty minutes of work in a fifty-minute period. Please report any error, ambiguity, out-of-coverage question, or item that takes much longer than its point value suggests.

Problem 1. (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Five points each: two for the circle, three for the reason.

- (a) **T** **F** The congruence $10x \equiv 6 \pmod{28}$ has exactly four solutions modulo 28.
- (b) **T** **F** Viewed as an element of S_9 , the permutation $(1\ 2\ 3\ 4)(5\ 6\ 7)$ has exactly two fixed points.
- (c) **T** **F** The rule $\varphi: \mathbb{Z}/8\mathbb{Z} \rightarrow \mathbb{Z}/12\mathbb{Z}$ given by $\varphi(\bar{x}) = \overline{2x}$ defines a group homomorphism.
- (d) **T** **F** The set of real 2×2 matrices of determinant 1 is a subgroup of $\mathrm{GL}_2(\mathbb{R})$.
- (e) **T** **F** If H and K are subgroups of a group G , then $H \cap K$ is a subgroup of G .
- (f) **T** **F** Every subgroup of a cyclic group is cyclic.

Problem 2. (25 points) **The unit group modulo 20.**

Work in the multiplicative group $U = (\mathbb{Z}/20\mathbb{Z})^\times$.

- (a) (4 points) List the elements of U .
- (b) (8 points) Find the orders of $\overline{3}$, $\overline{9}$, $\overline{11}$, and $\overline{19}$.
- (c) (5 points) Is U cyclic? Justify your answer.
- (d) (8 points) List every element of the subgroup $\langle \overline{3}, \overline{11} \rangle$. Is this subgroup all of U ?

Problem 3. (25 points) **Computing in D_{10} .**

Let

$$D_{10} = \langle r, s \mid r^5 = s^2 = 1, rs = sr^{-1} \rangle$$

be the dihedral group of order 10. Write every answer in the form r^i or sr^i , with $0 \leq i < 5$.

- (a) (6 points) Compute $(sr^2)(sr^4)$ and $(r^3s)^{-1}$.
- (b) (6 points) Determine the order of every element of D_{10} .
- (c) (7 points) Prove that $\langle sr, sr^3 \rangle = D_{10}$.
- (d) (6 points) List every subgroup of D_{10} and briefly justify that your list is complete.

Problem 4. (20 points) **Subgroups inside $\mathbb{Z}/24\mathbb{Z}$.**

Work in the additive group $\mathbb{Z}/24\mathbb{Z}$, and set

$$H = \{\bar{x} \in \mathbb{Z}/24\mathbb{Z} : 4\bar{x} = \bar{0}\}, \quad K = \langle \bar{8}, \bar{18} \rangle.$$

- (a) (7 points) Use the subgroup criterion to prove that H is a subgroup of $\mathbb{Z}/24\mathbb{Z}$.
- (b) (5 points) List the elements of H , give one generator of H , and determine all of its generators.
- (c) (8 points) Prove that $K = \langle \bar{2} \rangle$ by proving both containments. Then determine $H \cap K$.